

PRATEEK SINGH

B.Tech CSE, IIIT Guwahati (2023–2027) | Ps332927@gmail.com | 9473668934 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)
GSSoC 2026 Project Admin | GSSoC 2025 Contributor | Hacktoberfest 2025 | CodeSocial 2025

PROFESSIONAL SUMMARY

B.Tech Computer Science student at IIIT Guwahati specializing in AI-powered automation — building end-to-end workflows that integrate LLMs, APIs, and third-party services using n8n and Claude Code. Hands-on experience shipping AI-first products: an n8n-driven outreach automation platform with LLM-personalized messaging and real-time sentiment analysis, and a RAG-based enterprise analytics assistant. Proven ability to independently design, build, and ship reliable automation and AI applications, with additional experience in prompt engineering, agentic workflows, and remote cross-functional collaboration.

EDUCATION

Indian Institute of Information Technology, Guwahati

2023 – 2027

B.Tech in Computer Science & Engineering

TECHNICAL SKILLS

AI & Automation: n8n (workflow automation), Claude Code, LLM/API integration, Prompt Engineering, AI Agents, RAG Pipelines

Programming Languages: Java, Python, C++, JavaScript, TypeScript, HTML, CSS

Frontend: React.js (18/19), Next.js 14, TailwindCSS, Vite, Context API, CSS Modules

Backend: Node.js, Express.js, FastAPI, REST APIs, Servlet/JSP, JDBC

Databases: MySQL, MongoDB, Mongoose, Supabase, SQLite, Firebase (Auth & Firestore), ChromaDB

Machine Learning & NLP: PyTorch, Scikit-learn, GANs (Pix2Pix, U-Net, PatchGAN), BERTopic, NLTK VADER, Sentence-Transformers, LangChain, NumPy, Pandas

Tools & Platforms: Git, GitHub, VS Code, Linux, Jupyter Notebook, Ollama, Vercel, Google Places API

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Architecture, Secure Coding (OWASP)

PROJECTS

Lead-Scraper — AI-Powered B2B Lead Gen & Outreach Automation — [GitHub](#)

2026

React 19 • TypeScript • n8n • Supabase • Gemini LLM • Google Places API

- Built an end-to-end automation platform that discovers verified local business leads via the Google Places API and pushes them through n8n webhook pipelines to trigger multi-channel outreach.
- Designed n8n workflows integrating third-party APIs (WhatsApp, Gmail, LinkedIn) to automate the full outreach cycle — from lead discovery to personalized message dispatch — without manual intervention.
- Integrated the Gemini LLM to auto-generate personalized outreach emails and to power real-time AI sentiment and intent classification on inbound WhatsApp replies.
- Built an 8-stage CRM pipeline with Supabase real-time subscriptions, live conversion analytics, and quick-reply automation chips to optimize response workflows.

Automated Documentation Agent — [GitHub](#)

2026

n8n • Groq AI • Webhook Triggers • REST APIs

- Built a webhook-triggered workflow automation agent in n8n that uses Groq AI to auto-generate codebase documentation on trigger, removing a manual engineering task entirely.
- Demonstrated agentic automation by chaining AI inference, webhook handling, and data transformation into a single reliable pipeline without any custom backend code.

BizInsight AI — Enterprise Customer Feedback Analytics Platform — [GitHub](#)

2026

Next.js 14 • FastAPI • BERTopic • LangChain • ChromaDB • NLTK VADER • Sentence-Transformers

- Architected and shipped an enterprise feedback-analytics platform end-to-end, migrating a monolithic Streamlit app into a production-grade microservices system with a Next.js 14 frontend and a FastAPI backend.
- Automated sentiment analysis and unsupervised topic clustering (BERTopic, HDBSCAN, UMAP, Sentence-Transformers) to instantly sort negative reviews into business-relevant categories — Payment, Delivery, Technical, Product Quality — with live risk-threshold tracking.
- Built a Retrieval-Augmented Generation chatbot (LangChain, ChromaDB, OpenRouter Gemini) that answers business queries strictly from uploaded review data to eliminate hallucination, plus a public, auth-free RAG sandbox for prospective users.
- Engineered structured alerting for risk-level spikes and negative-review surges, enabling proactive business monitoring without manual review of raw data.

Sketch-to-Colorization using Conditional GAN — [GitHub](#)

2025

Python • PyTorch • U-Net • PatchGAN • Albumentations • Jupyter Notebook

- Built a conditional GAN (Pix2Pix-based) in PyTorch to automatically colorize black-and-white anime-style sketches, using a U-Net generator with skip connections to preserve fine sketch detail.
- Implemented a PatchGAN discriminator that evaluates local image patches instead of the whole image, producing sharper, more realistic textures in generated output.

EXPERIENCE & OPEN SOURCE

Project Admin — GSSoC 2026 (GirlScript Summer of Code)

2026

- Manage contributor onboarding, PR reviews, and issue triaging for an open-source project under GSSoC 2026, maintaining code quality standards and overall project health.
- Mentor incoming contributors and ensure timely merges, supporting a smooth and active contribution pipeline.

Research Collaborator — KathiraveluLab, University of Alaska Anchorage

2025 – 2026

- Contributed to GUARDA (Gateway for Uniform Access to Remote Data & Analytics), a federated database query platform, under Dr. Pradeeban Kathiravelu's research group, applying secure coding practices and structured issue tracking while collaborating remotely across time zones.
- Received an official reference letter from Dr. Kathiravelu (Assistant Professor of CS, UAA) recognizing contributions to research and open-source collaboration.

Open Source Contributor — GSSoC 2025, Hacktoberfest 2025, CodeSocial 2025

2025

- Merged multiple PRs across repositories (bug fixes, cleanup, docs, features) following OWASP guidelines and real-world Git workflows (branching, rebasing, CI checks).